

Prompt ID	Prompt Type
P1	Naive baseline
P2	Careful one-by-one counting
P3	Color-first decomposition
P4	Bounding box detection
P5	Self-verification recount
P6	Row-by-row structured counting
P7	Medication safety persona + uncertainty
P8	Describe scene first, then count
P9	Uncertainty-aware with confidence
P10	Numbered mental labeling

Full Prompt Text

How many pills are in this image?

Look at this image carefully. Count each individual pill one by one before giving me the final total number of pills.

First, identify each distinct pill color you see. Then count how many pills there are of each color. Finally, add those subtotals to give the total pill count.

Detect each pill in this image. For every pill, provide its approximate bounding box location as [top, left, bottom, right] and its color. Return the results as a list, then

Count all the pills in this image. Then recount them using a different method. If your two counts disagree, explain the discrepancy and give your best final answer.

Divide this image into a top half and bottom half. Count the pills in the top half first, then count the pills in the bottom half. Add the two numbers for the total count.

You are a medication adherence assistant helping an elderly patient verify their pills before they take them. Count every visible pill, identify each pill's color, and report your confidence (high, medium, or low). If any pills are touching, overlapping, or

First, describe everything you see in this image in detail, including the surface, the arrangement of the pills, their colors, and any pills that might be touching or

Count the pills in this image. For your answer, provide: (1) your count, (2) a confidence level (high/medium/low), and (3) explain what made counting difficult, if

Imagine you are placing a small numbered sticker on each pill in this image, starting with 1. For each pill, state its number, its color, and where it is in the image. After labeling all pills, report the highest number as your total count.

What It Tests

Control - no guidance at all

Whether slowing down helps (basic chain-of-thought)

Structured decomposition by color grouping, maps to pill verification use case

Uses Gemini's trained spatial detection to force commitment to specific pill regions before counting

Metacognition - can the model catch its own errors?

Spatial partitioning reduces estimation errors (supported by research)

Role-prompting + stakes framing + uncertainty flagging (addresses lit review gap)

Forces full visual processing before counting, prevents jumping to estimates

Self-assessment of confidence - key for safety-critical use

Inspired by Set-of-Mark (SoM) research - assigning individual IDs improves counting accuracy

Image ID	True Count	Pill Colors	Difficulty	Prompt ID	Prompt Type
IMG-01	7	green, 2 white, 2 brown	Easy	P1	Naive baseline
IMG-01	7	green, 2 white, 2 brown	Easy	P2	Careful one-by-one counting
IMG-01	7	green, 2 white, 2 brown	Easy	P3	Color-first decomposition
IMG-01	7	green, 2 white, 2 brown	Easy	P4	Bounding box detection
IMG-01	7	green, 2 white, 2 brown	Easy	P5	Self-verification recount
IMG-01	7	green, 2 white, 2 brown	Easy	P6	Row-by-row structured counting
IMG-01	7	green, 2 white, 2 brown	Easy	P7	Uncertainty-aware with confidence
IMG-01	7	green, 2 white, 2 brown	Easy	P8	Describe scene first, then count
IMG-01	7	green, 2 white, 2 brown	Easy	P9	Uncertainty-aware with confidence
IMG-01	7	green, 2 white, 2 brown	Easy	P10	Numbered mental labeling
IMG-02	7	green, 2 white, 2 brown	Medium	P1	Naive baseline
IMG-02	7	green, 2 white, 2 brown	Medium	P2	Careful one-by-one counting
IMG-02	7	green, 2 white, 2 brown	Medium	P3	Color-first decomposition
IMG-02	7	green, 2 white, 2 brown	Medium	P4	Bounding box detection
IMG-02	7	green, 2 white, 2 brown	Medium	P5	Self-verification recount
IMG-02	7	green, 2 white, 2 brown	Medium	P6	Row-by-row structured counting
IMG-02	7	green, 2 white, 2 brown	Medium	P7	Uncertainty-aware with confidence
IMG-02	7	green, 2 white, 2 brown	Medium	P8	Describe scene first, then count
IMG-02	7	green, 2 white, 2 brown	Medium	P9	Uncertainty-aware with confidence
IMG-02	7	green, 2 white, 2 brown	Medium	P10	Numbered mental labeling
IMG-03	7	green, 2 white, 2 brown	Hard	P1	Naive baseline
IMG-03	7	green, 2 white, 2 brown	Hard	P2	Careful one-by-one counting
IMG-03	7	green, 2 white, 2 brown	Hard	P3	Color-first decomposition
IMG-03	7	green, 2 white, 2 brown	Hard	P4	Bounding box detection
IMG-03	7	green, 2 white, 2 brown	Hard	P5	Self-verification recount
IMG-03	7	green, 2 white, 2 brown	Hard	P6	Row-by-row structured counting
IMG-03	7	green, 2 white, 2 brown	Hard	P7	Uncertainty-aware with confidence
IMG-03	7	green, 2 white, 2 brown	Hard	P8	Describe scene first, then count
IMG-03	7	green, 2 white, 2 brown	Hard	P9	Uncertainty-aware with confidence
IMG-03	7	green, 2 white, 2 brown	Hard	P10	Numbered mental labeling
IMG-04	15	green, 5 white, 5 brown	Hard	P1	Naive baseline
IMG-04	15	green, 5 white, 5 brown	Hard	P2	Careful one-by-one counting
IMG-04	15	green, 5 white, 5 brown	Hard	P3	Color-first decomposition
IMG-04	15	green, 5 white, 5 brown	Hard	P4	Bounding box detection
IMG-04	15	green, 5 white, 5 brown	Hard	P5	Self-verification recount
IMG-04	15	green, 5 white, 5 brown	Hard	P6	Row-by-row structured counting
IMG-04	15	green, 5 white, 5 brown	Hard	P7	Uncertainty-aware with confidence
IMG-04	15	green, 5 white, 5 brown	Hard	P8	Describe scene first, then count
IMG-04	15	green, 5 white, 5 brown	Hard	P9	Uncertainty-aware with confidence
IMG-04	15	green, 5 white, 5 brown	Hard	P10	Numbered mental labeling

[illegible]

Notes

Prompt ID	Prompt Type	# Correct (of 4)	Accuracy %	Mean Abs
P1	Naive baseline	4	100%	0.0
P2	Careful one-by-one counting	4	100%	0.0
P3	Color-first decomposition	4	100%	0.0
P4	Bounding box detection	4	100%	0.0
P5	Self-verification recount	4	100%	0.0
P6	Row-by-row structured counting	4	100%	0.0
P7	dication safety persona + uncertai	4	100%	0.0
P8	Describe scene first, then count	4	100%	0.0
P9	Uncertainty-aware with confidence	4	100%	0.0
P10	Numbered mental labeling	4	100%	0.0

[illegible]

Image ID	True Count	Pill Colors	Condition
IMG-01	7	3 green, 2 white, 2	Baseline/normal
IMG-02	7	3 green, 2 white, 2	Touching
IMG-03	7	3 green, 2 white, 2	Overlapping
IMG-04	15	5 green, 5 white, 5	Normal

Description	File Name
7 pills, clean background, good lighting, not touching	(fill in)
7 pills, some pills touching each other	(fill in)
7 pills, some pills overlapping — hardest condition	(fill in)
15 pills, normal conditions, high count challenge	(fill in)